

1] Explain the concept of SAVEPOINT in transaction management. How do ROLLBACK and COMMIT interact with savepoints?

Ans:

Concept of SAVEPOINT in Transaction Management:

In SQL, a SAVEPOINT is a point within a transaction that allows you to partially roll back to that specific point without affecting the entire transaction. It is used to mark intermediate stages within a transaction. If an error occurs or you need to undo certain operations, you can roll back to the SAVEPOINT rather than rolling back the entire transaction.

How ROLLBACK and COMMIT Interact with SAVEPOINT:

1. COMMIT:

A COMMIT statement finalizes all changes made during a transaction, including those made after any SAVEPOINTS. Once a transaction is committed, all changes are permanently saved to the database.

Interaction with SAVEPOINT: Once a COMMIT is issued, all SAVEPOINTS within the transaction are cleared. You cannot roll back to a SAVEPOINT after committing the transaction.

2.ROLLBACK:

A ROLLBACK statement undoes all changes made during a transaction up to the point of the last commit or to a specific SAVEPOINT.

Interaction with SAVEPOINT: If you issue a ROLLBACK TO SAVEPOINT <savepoint_name>, the transaction will revert to the state it was in when the specified SAVEPOINT was created. However, if you issue a full

ROLLBACK without specifying a SAVEPOINT, the entire transaction will be rolled back, including all SAVEPOINTS.

Example:

Start a new transaction

BEGIN;

Insert some records into a table

INSERT INTO employees (employee_id, employee_name, salary) VALUES
(1, 'John Doe', 50000);

Set a SAVEPOINT after the first insert

SAVEPOINT before_update;

Update a record (let's assume this might fail later)

UPDATE employees SET salary = 55000 WHERE employee_id = 1;

Check for error and decide to roll back to the savepoint if needed

In case of an error or decision to undo the update, we can use ROLLBACK
TO SAVEPOINT

ROLLBACK TO SAVEPOINT before_update;

Continue with more operations or a commit

COMMIT the transaction to make all changes permanent

COMMIT;